

- A** 1 and 3 **B** 2 and 4 **C** 3 only **D** 4 only

- 11** Blood is normally slightly alkaline and maintained in the pH range 7.35–7.45. During exercise lactic acid is produced.

Which reaction occurs during exercise to maintain the pH of the blood?

- A** $\text{H}_2\text{CO}_3 + \text{OH}^- \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$
- B** $\text{HCO}_3^- + \text{OH}^- \rightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$
- C** $\text{HCO}_3^- + \text{H}_3\text{O}^+ \rightarrow \text{H}_2\text{CO}_3 + \text{H}_2\text{O}$
- D** $\text{CO}_3^{2-} + \text{H}_3\text{O}^+ \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$

- 12** Some information about some of the elements of Period 5 of the Periodic Table is shown.

element	Rb	Sr	In	Sn	Sb	Te	I	Xe
electrical conductivity	high	high	high	high	high	semi-conductor	zero	zero